CTS-BRK-0117 rev C

FR-330V Front Brake Rotor | Component Technical Specification

ILLUSTRATIVE DOCUMENT — NOT A REAL PROGRAM This specification is **fictional**, written to demonstrate an agentic CAE workflow. The part, the programme, the document number and the vehicle do not exist and correspond to no manufacturer's product. Nothing here is derived from, or contains, any company's proprietary or confidential information. The format follows real automotive practice so the demonstration is meaningful, and the thermal limits are sourced from open literature (§3.2). It is **not** engineering guidance and must not be used to design a real brake: a production rotor is sized by repeated-stop duty cycles, not by the single stop analysed here (§6).

FR-330V Front Brake Rotor — Performance Variant

Document	CTS-BRK-0117 rev C
Program	S-Platform MY28 Performance Pack *(fictional)*
Part	Front brake rotor, vented, Ø330
APQP phase	Phase 2 — Product Design & Development
Owner	Chassis / Foundation Brake
Status	Released for design *(illustrative — see banner)*
Supersedes	CTS-BRK-0117 rev B (Ø320 solid, carryover)

Document type. This is a **component technical specification** — the bottom tier of the automotive requirements cascade (**vehicle** → **subsystem** → **component**) — paired with a **DVP&R extract** (Section 5) naming how each requirement is proven. The requirements below are numbered, individually verifiable, and each carries the method that closes it. **Naming varies by OEM and there is no standards-body term for the document itself.** GM calls this tier a Component Technical Specification (CTS), under a Subsystem Technical Specification (SSTS) and a Vehicle Technical Specification (VTS). Ford's equivalents are the System Design Specification (SDS) and Engineering Specification (ES); Stellantis issues a Statement of Requirements (SoR); VW Group and other German OEMs use a component *Lastenheft*, rendered officially in English as a performance specification. IATF 16949 §8.3.3.1 mandates the **content** — "product design input requirements" — and deliberately names no carrier document; AIAG PPAP §2.2.1 calls the resulting controlled artefact the **design record**. What this document is **not** is a Product Requirements Document. "PRD" does not appear anywhere in IATF 16949 or the AIAG core manuals; it is software, consumer-product and medical-device vocabulary.

1. Scope and application

The S-Platform currently ships a Ø320 **solid** front rotor carried over from the outgoing car. It is adequate for the base powertrain but has been the limiting component in two places:

1. Press fade testing on the base car reported pedal lengthening on the third

consecutive high-speed stop.

2. The Performance Pack adds 90 kW and 140 kg over base, which the carryover

foundation brake was never sized for.

This specification releases a **vented, cross-drilled Ø330 rotor** for the Performance Pack only. Base trim retains the carryover part; there is no common-part requirement between them beyond the hub interface.

The hub interface is frozen. The wheel bearing, knuckle and wheel are all carryover. Any change to PCD, centre bore or mounting-face offset re-opens three other subsystems and is out of scope for this program.

2. Vehicle and duty cycle

These numbers are the design point. Everything downstream traces to them.

Parameter	Value	Source
Kerb mass + driver	**1800 kg**	Vehicle mass study VMS-114
Design entry speed	**100 km/h**	Worst routine stop, customer usage study
Design deceleration	**0.8 g** (7.848 m/s²)	Severest *routine* stop; tyres reach 1.0–1.2 g
Front axle bias	**0.70**	Static + dynamic weight transfer at 0.8 g
Tyre rolling radius	**0.33 m**	245/40 R19
Front rotors	2	

Derived duty, per front rotor — the single stop this program is sized around:

Quantity	Value
Stop duration	3.54 s
Stopping distance	49.2 m
Energy absorbed	**243 kJ**
Mean power	**68.6 kW**
Wheel speed at entry	84.2 rad/s (804 rpm)

> **Design point rationale.** 0.8 g is not the tyre limit — it is the severest stop a normal driver produces repeatedly. Sizing to 1.2 g would add unsprung mass for a condition customers reach a handful of times in a vehicle life; sizing to 0.5 g would fail the press fade test that started this program.

3. Requirements

M = mandatory (gate blocker) · **S** = should · **C** = could

3.1 Package and interface

ID	Req	Value	Class	Verification
PKG-01	Outer diameter	330 mm max	M	CAD measure
PKG-02	Overall thickness	32 mm nominal	M	CAD measure
PKG-03	Bolt circle diameter	**115 mm (5 × Ø14)** — carryover, frozen	M	CAD measure
PKG-04	Centre bore	**Ø70** — carryover, frozen	M	CAD measure
PKG-05	Mounting face offset	+43 mm from rotor mid-plane	M	CAD measure
PKG-06	Wheel clearance	≥ 6 mm radial to 19″ rim inner	S	Package check

3.2 Thermal — the reason this program exists

ID	Req	Value	Class	Verification
THM-01	Peak rotor **friction-surface** temperature, single design stop	≤ 923 K (650 °C)	M	**CFD**
THM-02	Peak pad friction-surface temperature, single design stop	≤ 623 K (350 °C)	M	CFD
THM-03	Rotor bulk temperature rise, single design stop	≤ 60 K	S	CFD
THM-04	No rotor surface region above 1143 K (870 °C)	—	M	CFD
THM-05	Vent passages must contribute measurable cooling	modelled explicitly	M	CFD, distinct vent HTC
THM-06	Caliper / brake fluid temperature	≤ 428 K (155 °C)	M	**Deferred — see §6**

Basis of the thermal limits. These are not round numbers picked for convenience; each has a named failure mode:

ID	Failure mode it guards against	Where the number comes from
THM-01	Heat checking, hot judder, thermo-mechanical fatigue	Industry **Single-Stop Temperature Rise (SSTR)** design practice: 350–400 °C drum, 550 °C solid disc, **600–650 °C vented disc**. This part is a vented disc, so it takes the top band.
THM-02	Fade — phenolic binder decomposition and outgassing in the friction material	First-stage thermal fade for organic and low-metallic compounds is consistently reported at ~350 °C
THM-04	Loss of iron material capability	Grey cast iron is reported serviceable to ~870 °C. **Between THM-01 and THM-04 the mechanism is fatigue and heat checking, not yield** — exceeding 650 °C degrades life, it does not fail the part in one stop
THM-06	Vapour lock — fluid boils, pedal goes long	DOT 4 minimum **wet** boiling point, 155 °C. Alpine-descent fluid temperatures of 150–200 °C are documented, so this is a live constraint, not a formality

SSTR is a temperature *rise*, not an absolute. THM-01 is written as an absolute because the design stop starts from an initial brake temperature of 300 K; rise and absolute differ by 27 K here. On a duty cycle with a hot starting temperature they do not, and the limit must be re-derived.

THM-01 and THM-02 are the gate, and THM-02 is expected to govern. Pad compound conductivity is ~40× lower than iron, so the pad runs hotter against a limit 300 K lower. A review that concludes "the rotor is fine" has answered the wrong question.

This specification sizes a SINGLE STOP. It is not a duty cycle. Repeated stops (AMS 10-stop, Alpine descent, autobahn stop) drive the thermal design of a production rotor and are **not** covered here — see §6.

3.3 Mass and geometry

ID	Req	Value	Class	Verification
MAS-01	Rotor mass	≤ 13.5 kg	M	CAD mass properties
GEO-01	Vented construction, two cheeks + vanes	—	M	CAD
GEO-02	Cheek thickness	10 mm each	S	CAD
GEO-03	Vent gap	12 mm	S	CAD
GEO-04	Curved (directional) vanes	≥ 18 vanes	S	CAD
GEO-05	Cross-drilled cooling holes through the swept band	≥ 30 holes	C	CAD
GEO-06	Hat attaches to the **inboard** cheek, vent inlet unobstructed	—	M	CAD section

GEO-06 is not cosmetic. A hat that spans the full height at the ring bore blocks the vent inlet, and the cooling result is then quietly wrong. Reference teardown of a comparable Ø380 rotor confirms the web lands on the inboard cheek. See `docs/HAT_ATTACHMENT_EVIDENCE.md`.

3.4 Friction pair

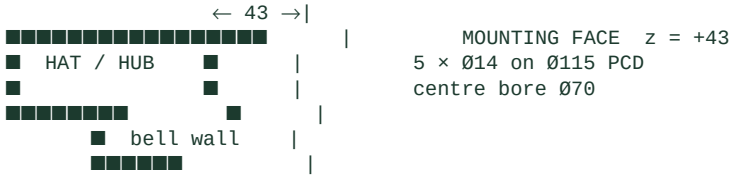
ID	Req	Value	Class	Verification
PAD-01	Pad sector angle	60°	M	CAD
PAD-02	Pad radial span	r 120–160 mm	M	CAD
PAD-03	Pad thickness (friction material)	12 mm	S	CAD
PAD-04	Backing plate	6 mm steel, r 115–165	S	CAD
PAD-05	Caliper must clear the full pad stack	slot ≥ stack + 4 mm	M	CAD interference

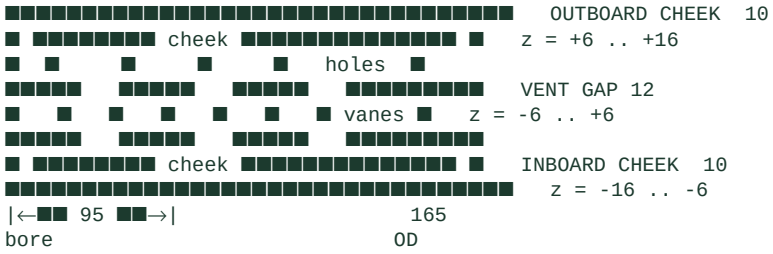
3.5 Materials

ID	Component	Material	ρ	c_p	k
MAT-01	Rotor	Grey cast iron	7200	447	52
MAT-02	Pad	Low-metallic compound	2500	900	1.2
MAT-03	Backing plate	Steel	8030	502	16.3

4. Concept sketch

See `concept_sketch.png`. Meridional section, dimensions in millimetres.





Vanes are **curved**, swept ~32° from bore to OD, running the same direction as forward rotation so they pump outward. Cooling-hole rows follow the vane spiral, offset half a pitch so a hole never breaks into a vane.

5. Design Verification Plan & Report (DVP&R) — extract

ID	Requirement	Method	Stage	Acceptance
DV-01	PKG-01..05	CAD measurement	Design	Within tolerance
DV-02	MAS-01	CAD mass properties, cross-checked against closed-form volume	Design	≤ 13.5 kg, arithmetic agrees within 1%
DV-03	GEO-06	CAD section view	Design	Vent inlet open
DV-04	THM-01..05	**Transient conduction CFD**, rotating rotor, vehicle-derived duty	Analysis	Section 3.2 limits
DV-05	Energy conservation of the CFD model	Injected power vs vehicle kinetic energy	Analysis	Within ±2%
DV-06	PAD-05	CAD interference check	Design	Zero interference
DV-07	Fade resistance, repeated stops	Dyno, AK-Master	Validation	**Not in this phase**
DV-08	Thermal cracking, thermal shock	Dyno	Validation	**Not in this phase**
DV-09	THM-06 caliper / fluid temperature	Thermal model including caliper, or dyno thermocouple	Analysis + Test	**Not in this phase** — the caliper is drawn but deliberately excluded from the thermal model (§6)
DV-10	Repeated-stop duty cycle (AMS 10-stop, Alpine descent)	Transient thermal, multi-event	Analysis	**Not in this phase** — see §6

DV-04 is what this analysis delivers. DV-07 through DV-10 need physical test, a caliper in the thermal model, or a repeated-stop duty cycle. A single-stop conduction model cannot answer them and must not be presented as if it can.

6. Out of scope

Explicitly excluded from this phase, and from the analysis that supports it:

- Structural stress, coning, thermal-crack prediction
- Pad wear, transfer-film stability
- Fade from binder outgassing

- Disc thickness variation / judder
- Airflow solution — the convection coefficients are **inputs**, not solved
- NVH, corrosion, manufacturing process design (APQP Phase 3)

Two exclusions worth naming explicitly, because a brake engineer will look for them and their absence is a scope decision rather than an oversight:

repeated events — the AMS 10-stop press evaluation (10 consecutive stops from 100 km/h at GVW), Alpine descent (conventionally ~10 miles at 10% gradient), and the autobahn stop — not by one stop from 100 km/h. Disc temperatures in those events reach 600–700 °C, where this single stop does not. **This specification sizes the single stop only** (DV-10).

Alpine-descent fluid temperatures of 150–200 °C sit at or above the DOT 4 minimum wet boiling point of 155 °C, and soak-back after the vehicle stops is the governing case. The caliper is drawn in CAD but **suppressed before meshing**, so this model cannot answer it (DV-09).

- **Repeated-stop duty cycles.** A production rotor is thermally sized by
- **Caliper and brake fluid temperature.** THM-06 is a real constraint —

7. Gate criteria — Phase 2 exit

Phase 2 exits to Phase 3 when:

1. All **M** requirements verified or dispositioned with a written rationale

2. DV-01..DV-06 complete with recorded evidence

3. Design review package issued: drawing, 3D model, CFD report, recommendations

4. Open items carry an owner and a target phase

8. Traceability to the analysis inputs

Every number the CFD uses comes from Section 2, not from a solver example. This is the check that stops an inherited parameter set producing a confident answer for the wrong vehicle.

CTS	Analysis input	Value
§2 mass, speed, decel, bias	`q_dot`	2.93e9 W/m³
§2 speed, tyre radius	`omega`	–56.12 rad/s (energy-weighted mean)
§2 decel	`n_steps`	354 at 0.01 s
§3.2 THM-05	`htc_vent`	130 W/m²K, distinct from outer
§3.5	Per-zone materials	MAT-01..03

`solve.py` recomputes `q_dot`, `omega` and `n_steps` from the vehicle block and **fails the run** if the stated values disagree by more than 1%.